

Soyeb Pervez Jim

Software Engineer

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Professional Summary

Software engineer with remote experience building production LLM agents, semantic retrieval, and multi-tenant SaaS for e-commerce. Co-author on two Elsevier journal papers applying physics-informed deep learning to industrial systems; author of C++/Qt desktop software with 313 GitHub stars. English at CEFR C1.

Technical Skills

Languages: TypeScript, Python, C++, QML, JavaScript, Swift, Lua, Bash

AI/ML: LLM agents, tool calling, retrieval-augmented generation (RAG), vector/semantic search, embeddings, AI SDK, OpenAI, Anthropic and Google models, PyTorch, ONNX, scikit-learn, CNN/TCN, biosignal processing

Web/Systems: Next.js, React, Node.js, Express, Sequelize, MySQL/Vitess, server-sent events, Stripe, Astro, Qt 6, QML, Wayland, Linux, Nix

Professional Experience

Software Engineer

July 2025 – Present

Fuego.io

Remote

- Lead contributor to the tool-calling chat assistant behind a multi-tenant Shopify commerce platform: streaming responses, product search, recommendations, and reviews as structured tool results.
- Built semantic product search over live catalogs, with a scheduled re-embedding job keeping every tenant's index current.
- Automated per-brand context generation, so a new tenant gets a customer-care knowledge base from its own site with no manual authoring.
- Own the admin surface and metering: analytics, token-usage tracking, and Stripe billing across tenants.

Selected Projects

HyprFM

C++, Qt 6, QML, Wayland

- Keyboard-first file manager with Miller columns, split panes, async transfers, device mounting, and live-reloading TOML themes; QML presentation split from C++ backend services, shipped on AUR, Flatpak, AppImage, and Nix. 313 stars and an independent [LinuxLinks review](#).

Neural Network X-Ray

Next.js, React, TypeScript, ONNX Runtime

- Browser visualizer showing activations across all 13 CNN layers as users draw Latin or Bengali characters, inference running locally through WebAssembly. Trained the 146-class model on roughly 980,000 EMNIST and BanglaLekha images, published on [Hugging Face](#).

Research & Publications

Physics-Informed Multi-Task Neural Network for Leak Detection in Multiphase Pipelines *Published, 2026*
Process Safety and Environmental Protection (Elsevier) | doi.org/10.1016/j.psep.2026.109609

- PCE-PINN, a multi-task network predicting leak presence, diameter, and location from inlet/outlet pressure and flow alone: $F1 = 1.00$ detection, $R^2 = 0.906$ and 0.941 , a 16.5% location gain over the best classical baseline.

Physics-Informed Embed-to-Control with Per-Producer MLP Heads for CO₂ Storage Surrogate Modelling
Fuel Processing Technology (Elsevier), submitted 2026 *Under review*

- Replaced the shared linear well model in an E2CO reservoir surrogate with per-producer MLP heads, trained under PINN residuals on the simulator's own finite-volume operator: saturation RMSE -38% , pressure RMSE -35% , per-well R^2 from 0.82 to 0.91 in 16 min on one T4.

Challenge papers. AI4Pain 2026, ACII Grand Challenge (under review): two-stage physiological pain classifier, 0.572 test accuracy against a 0.398 baseline. MUMIDC, ABC2026: CNN2D-TCN over EEG, EMG, and IMU, 87.9% macro F1 under 16-fold mixed LOSO.

Education & Achievements

BSc in Electrical and Electronic Engineering, University of Dhaka

2021 – Present, Final Year

Final-year thesis on abnormal activity recognition | [EF SET English certificate: C1](#)

Competitive programming. [Codeforces](#) peak 1714 (Expert), 769 problems solved over 121 rated rounds. ICPC Dhaka Regional finalist twice: 37th/308 (2024), 37th/225 (2023). [National Collegiate Programming Contest](#) 27th/196; Meta Hacker Cup round 2 in 2023 and 2024.

Olympiad. Bangladesh Mathematical Olympiad national first runner-up (2017) and second runner-up (2016); co-authored [Ganit Olympiader Hatekhori](#), an olympiad primer (Swapno '71, 2024). Published [worked olympiad solutions](#) to 6.5K followers.